

Carl Andrew Seelhoff

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Education

University of Michigan, Ann Arbor, MI

August 2023 – Present

PhD Candidate in Mechanical Engineering

Dissertation Title (tentative): Towards Translation of Continuous Biomimetic Control onto Commercial Powered Knee Prostheses

Department of Mechanical Engineering | Rackham Merit Fellow

Expected Graduation: Fall 2027 (M.S. Robotics), Fall 2028 (PhD Mechanical Engineering)

Advisor: Robert Gregg, PhD

GPA: 3.851/4.0

Massachusetts Institute of Technology, Cambridge, MA

August 2018 – February 2023

Bachelor of Science in Mechanical Engineering and minor in Statistics and Data Science

Undergraduate Thesis: [*Evaluating a reflexive neuromuscular gait model against real-world results*](#)

GPA: 4.8/5.0

Experience

Mechanical Engineering Intern | *Otto Bock Healthcare LP – Salt Lake City, UT* **January – July 2023**

- Performed pilot experiments for novel prosthetic liner cleaning technology
- Developed a standalone application in MATLAB to automate processing of routine test data for prosthesis liner coupons, allowing real-time visualization for SPC

Researcher on the Assistive Technology Provision and Access Team

June 2019 – January 2023

Biomechatronics Group | MIT Media Lab – Cambridge, MA

- Analyzed and compiled information on prosthetic care and services, barriers and mitigation strategies across the United States and internationally in several countries.
- Developed a condition-specific metric to capture the value of emerging technologies and better direct their development.
- Applied Digital Imaging Correlation methods to produce computational models of limbs for the development of prosthetic sockets that improve patient comfort and quality of care.

Researcher on the Health Delivery and Health Equity Team

June 2019 – January 2023

Biomechatronics Group | MIT Media Lab – Cambridge, MA

- Contributed to the design of an ADA-compliant mobile vehicle capable of extending prosthetic and diabetic services to remote, rural and marginalized communities.
- Produced design iterations based on logistical and engineering restraints and requirements incorporating appropriate technological methods.
- Created blueprints and photorealistic renders of several concept models to demonstrate to donors, contributing to the successful funding injection of \$24M towards the creation of K. Lisa Yang Center for Bionics at MIT.

Advocacy

Certified Lead Advocate, Amputee Coalition of America – Washington D.C.

2025-Present

- Attended one-month workshop on grassroots policy skills needed to lead statewide push for passing of So Every BODY Can Move (SEBCM) Legislation
- Preparing to assist with introduction and enactment of SEBCM for Michigan

Lobbyist, American Orthotic and Prosthetic Association – Washington D.C.

2015-2019

- Annually attended summit for O&P insurance advocacy on Capitol Hill
- Met with members of Congress to lobby for insurance fairness for amputees
- Advocated for research and development for pediatric prostheses

Peer-Reviewed Journal Articles

1. T. K. Best, **C. A. Seelhoff**, J. Wensman, and R. Gregg, “The clinical effects of the Össur Power Knee with phase-based and default control during sitting, standing, and walking,” *Journal of NeuroEngineering and Rehabilitation (JNER)*, 2025.

Peer-Reviewed Conference Proceedings

1. **C. A. Seelhoff**, T. K. Best, and R. D. Gregg. “Adapting Biomimetic Kinematics for Controlling a Powered-Knee, Passive-Ankle Prosthesis Across Inclines,” *IEEE Int Conf Rehab Robotics (ICORR)*, 2025.
2. T. K. Best, **C. A. Seelhoff**, and R. D. Gregg. “Implementation and Validation of a Data-Driven Variable Impedance Controller on the Össur Power Knee,” *IEEE Int Conf Rehab Robotics (ICORR)*, 2025.

Conference Presentations & Speaking Opportunities

1. **C. A. Seelhoff**, C. A. Laubscher, T. K. Best, and R. D. Gregg. Adaptation of Continuous Biomimetic Control to Powered-Knee, Passive-Ankle Prostheses Across Inclines and Speeds. Dynamic Walking 2026, Seattle, WA, USA, June 2026. *Podium presentation.*
2. **C. A. Seelhoff**, A leg up: lessons learned as an amputee in powered prosthesis research. 4th Annual Disability Visibility in Engineering Symposium, Ann Arbor, MI, USA, April 2026. *Student Keynote Speaker.*
3. **C. A. Seelhoff**, T. K. Best, and R. D. Gregg. Adapting Biomimetic Kinematics for Controlling a Powered-Knee, Passive-Ankle Prosthesis Across Inclines. 19th IEEE/RAS-EMBS International Conference on Rehabilitation Robotics (ICORR 2025), Chicago, IL, USA, May 2025. *Podium Presentation.*
4. A. Shah, **A. Seelhoff**, F. Riccio-Ackerman, H. Herr. Designing for Democratizing Prosthetic and Diabetic Care: *A Resilient Model for Healthcare Delivery*. 16th Annual Mobile Health Clinics Conference, September 2020. *Virtual.*

Skills

- Machines: Mill/lathe, surface-mount soldering, Injection molding, 3-axis CNC machining (HAAS)
- Programming: Python 3 (intermediate), MATLAB (advanced) + Simulink (beginner), ROS (beginner), C++ (intermediate), git (intermediate)

- CAD/CAM: Solidworks (intermediate), Autodesk Fusion (intermediate), SketchUp (intermediate)
- Motion Capture Software: VICON Nexus (intermediate)
- Biomechanics Software: OpenSim (beginner)
- Rendering: Solidworks Visualize, V-Ray for SketchUp (intermediate)
- Media Editing: Adobe Premiere Pro (beginner)
- Languages: English (native), German (conversational)

Awards

- **NSF GRFP Awardee** (2025) \$169,000
- Rackham Merit Fellowship (2023)
- California Heroes Award (2017)
 - Presented by State Senator Joel Anderson for lobbying efforts in Washington, D.C.